

Summary



On 3rd of December 2025, Universiti Teknologi Malaysia (UTM) conducted an industrial visit to the Universiti Tun Hussein Onn Malaysia (UTHM) to bridge the gap between classroom theory and real-world enterprise system design. The focus of the visit is to understand the data centre infrastructure of the host university and its strategic methods for digital transformation. The students visited the server facilities and had a detailed Q&A session with the IT management team. They learned the balance between large educational institutions managing legacy hardware, data security and modern user needs.

Technologies and Issues Discussed

The technical brief shows that the architecture of UTHM prioritises control and redundancy rather than the purely cloud-based trend. A key highlight is the adoption of a Tier 3 containment data center layout, which create isolated zones for hot exhaust air or cold supply air that improving cooling efficiency. Additionally, the infrastructure uses a robust Disaster Recovery Process to safeguard data integrity. Unlike many modern enterprises that have turned to cloud computing, UTHM maintains a local server ecosystem to ensure data sovereignty.

For data integration, the admin explained that they used a master database as a single source of student data. Most of the systems obtain data from this master data source through API, while the learning management system (LMS) operates as an independent entity, highlighting the challenge of integrating different systems. Finally, the team emphasized that their digital transformation is driven by the goal of enhancing user experience. In order to achieve this, they follow the "phase-to-phase" migration strategy to ensure the stability of users during the update.

Reflection

I learned that a powerful system depends on hardwares and strategies. By looking the "hot" and "cold" server, I learned how an organisation can balance the cost and security to prevent downtime. I also learned that digital transformation is not just create a new technology; it is about to improve user experience through careful and phase to phase migration. This experience emphasises how important to have a great planning before implementing a new system.

I get to understand how a real data centre operates and why it is important for student learning. I saw how their IT team is divided into Infrastructure and Information groups, each managing different parts of the system. Although they explained their Enterprise Architecture only at a basic level, it still showed how structured planning guides their operations. I also learned about their phase-by-phase migration strategy for moving systems into the new data centre.

I thought it was important for us as students to not only see the information we learn applied in practice but also be able to raise questions and talk with the people working with these systems. The guided tour was a nice addition to the presentations as we got an exclusive opportunity to see a real data center, soon to be in action. I also found it informative to hear about how much preparation and work is required for digital transformations and how much time and resources it takes.

I discovered the importance of technical planning and good management in digital transformation. Enhancing user experience and to keep updating current technology are the significance of digital transformation. I also realized that physical infrastructure is important in system reliability. The different types of aisle setup in the data center is interesting for me as I get to know its main purpose which is power saving by enhancing the cooling efficiency.

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